

Package ‘riskweightedassets’

September 14, 2026

Type Package

Title Reproducible Risk-Weighted Asset Calculations

Version 1.0.0

Description Provides transparent, deterministic and auditable calculations of risk-weighted assets, own-funds requirements, interest-rate risk in the banking book and related capital metrics. It supports canonical in-memory tables and versioned spreadsheet datasets, strict validation, synthetic reference profiles, bitemporal snapshots, calculation controls and traceable regulatory source metadata. The implementation is intended for analytical, educational and model-validation use and does not constitute legal or supervisory advice.

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URL <https://github.com/rds0001/risk-weighted-assets-r>,
<https://riskdatascience.net/impressum/>,
<https://riskdatascience.net/datenschutzerklaerung/>

BugReports <https://github.com/rds0001/risk-weighted-assets-r/issues>

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Depends R (>= 4.1.0)

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utils,
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rmarkdown,
testthat (>= 3.0.0)

Config/testthat/edition 3

Roxygen list(markdown = TRUE)

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VignetteBuilder knitr

R topics documented:

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as.data.frame.rwa_validation_report
<i>Convert a Validation Report to a Data Frame</i>

Description

Convert a Validation Report to a Data Frame

Usage

```
## S3 method for class 'rwa_validation_report'
as.data.frame(x, row.names = NULL, optional = FALSE, ...)
```

Arguments

x	An rwa_validation_report.
row.names	Unused compatibility argument.
optional	Unused compatibility argument.
...	Unused.

Value

A data frame with one row per structured validation message.

calculate_dataset	<i>Calculate a Canonical Workbook Dataset</i>
-------------------	---

Description

Validates a dataset, creates an immutable deterministic run directory and writes six structured output workbooks plus a JSON run manifest.

Usage

```
calculate_dataset(dataset)
```

Arguments

dataset	Dataset directory containing an inputs subdirectory.
---------	--

Details

Output is written below dataset/outputs/<run-id>. The run ID fingerprints the input files, code and package version.

Value

An rwa_calculation_result object with output paths.

See Also

[validate_dataset\(\)](#), [generate_synthetic_dataset\(\)](#)

Examples

```
root <- file.path(tempdir(), "rwa-calculation-example")
dataset <- generate_synthetic_dataset(root, bank_profile = "KSA_BANK")
calculate_dataset(dataset)
```

calculate_tables	<i>Calculate Canonical In-Memory Tables</i>
------------------	---

Description

Validates and calculates canonical tables without producing spreadsheet output. A parallel fully-loaded result is calculated from the same snapshot.

Usage

```
calculate_tables(tables, run_id = NULL, project_root = NULL)
```

Arguments

tables	Named list of canonical data frames.
run_id	Optional caller-defined run identifier.
project_root	Optional root used to resolve local source documents.

Details

The return includes applied and fully-loaded result tables. All calculations use one official bitemporal snapshot. The function does not write files and does not change the global random-number stream.

Value

An `rwa_calculation_result` object.

See Also

[generate_synthetic_tables\(\)](#), [calculate_dataset\(\)](#)

Examples

```
tables <- generate_synthetic_tables(bank_profile = "KSA_BANK")
result <- calculate_tables(tables)
result$metrics[c("RWEA_KSA", "TREA")]
```

create_workspace

Create a Writable RWA Workspace

Description

Exports the package's synthetic reference environment to a caller-controlled directory. Existing non-empty directories require explicit overwrite.

Usage

```
create_workspace(path, overwrite = FALSE)
```

Arguments

path	Destination directory.
overwrite	Whether existing files may be replaced.

Value

An `rwa_workspace` object.

Examples

```
workspace <- create_workspace(file.path(tempdir(), "rwa-example"), overwrite = TRUE)
print(workspace)
```

default_workspace	<i>Default Writable Workspace</i>
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Description

Uses RWA_WORKSPACE when it is set, otherwise a rwa-workspace directory below the current working directory. The function only describes the path; it does not create or modify it.

Usage

```
default_workspace()
```

Value

An rwa_workspace object.

Examples

```
default_workspace()
```

generate_synthetic_dataset	<i>Generate a Synthetic Workbook Dataset</i>
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Description

Generate a Synthetic Workbook Dataset

Usage

```
generate_synthetic_dataset(
  root,
  as_of = NULL,
  version = NULL,
  seed = NULL,
  bank_profile = "MID_SIZE_UNIVERSAL",
  config_root = NULL,
  overwrite = FALSE
)
```

Arguments

root	Parent directory for dated datasets.
as_of	Reporting date.
version	Dataset version.
seed	Integer lineage seed.
bank_profile	Profile identifier.
config_root	Optional profile configuration directory.
overwrite	Whether an existing dataset may be replaced.

Details

The function writes 16 canonical workbooks plus a dataset manifest. An existing non-empty target requires `overwrite = TRUE`.

Value

The generated dataset directory.

See Also

[validate_dataset\(\)](#), [calculate_dataset\(\)](#)

Examples

```
generate_synthetic_dataset(file.path(tempdir(), "rwa-runs"),
                           bank_profile = "KSA_BANK")
```

generate_synthetic_tables

Generate Synthetic Canonical Tables

Description

Materializes a complete synthetic banking profile from immutable package fixtures. The seed is recorded for lineage; no business distributions are generated at runtime.

Usage

```
generate_synthetic_tables(
  as_of = NULL,
  seed = NULL,
  bank_profile = "MID_SIZE_UNIVERSAL",
  config_root = NULL
)
```

Arguments

as_of	Reporting date. Defaults to the profile base date.
seed	Integer lineage seed.
bank_profile	Profile identifier.
config_root	Optional directory containing profile YAML files.

Value

A named list of canonical data frames.

See Also

[calculate_tables\(\)](#), [generate_synthetic_dataset\(\)](#)

Examples

```
tables <- generate_synthetic_tables(bank_profile = "KSA_BANK")
length(tables)
```

list_reference_datasets

List Bundled Reference Datasets

Description

List Bundled Reference Datasets

Usage

```
list_reference_datasets()
```

Value

A data frame with dataset identifiers and versions.

Examples

```
list_reference_datasets()
```

`list_reference_profiles`*List Bundled Reference Profiles*

Description

Returns metadata for every synthetic reference profile included with the package. No customer or production data is included.

Usage

```
list_reference_profiles()
```

Value

A data frame with profile identifiers, versions, base dates, seeds, default dataset versions and descriptions.

Examples

```
list_reference_profiles()
```

`print.rwa_calculation_result`*Print a Calculation Result*

Description

Print a Calculation Result

Usage

```
## S3 method for class 'rwa_calculation_result'  
print(x, ...)
```

Arguments

<code>x</code>	An <code>rwa_calculation_result</code> .
<code>...</code>	Unused.

Value

`x`, invisibly.

```
print.rwa_validation_report
```

Print a Validation Report

Description

Print a Validation Report

Usage

```
## S3 method for class 'rwa_validation_report'  
print(x, ...)
```

Arguments

x	An rwa_validation_report.
...	Unused.

Value

x, invisibly.

```
print.rwa_workspace
```

Print a Workspace

Description

Print a Workspace

Usage

```
## S3 method for class 'rwa_workspace'  
print(x, ...)
```

Arguments

x	An rwa_workspace.
...	Unused.

Value

x, invisibly.

regulatory_sources	<i>Regulatory Source Metadata</i>
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Description

Returns links and archived checksums for official sources used to design the engine. The source documents themselves are deliberately not redistributed.

Usage

```
regulatory_sources()
```

Value

A data frame. The redistributed column is always false for the bundled metadata.

Examples

```
regulatory_sources()
```

validate_dataset	<i>Validate a Canonical Dataset</i>
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Description

Reads and validates all canonical input workbooks without calculating or writing any output files.

Usage

```
validate_dataset(dataset)
```

Arguments

dataset	Dataset directory containing an inputs subdirectory.
---------	--

Value

An rwa_validation_report object.

See Also

[calculate_dataset\(\)](#), [generate_synthetic_dataset\(\)](#)

Examples

```
root <- file.path(tempdir(), "rwa-validation-example")
dataset <- generate_synthetic_dataset(root, bank_profile = "KSA_BANK")
as.data.frame(validate_dataset(dataset))
```

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